

FRANCESCO GRUSSU

CONTACT AND ONLINE PROFILES

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EDUCATION AND TRAINING

PhD in MR Physics	2012-2016	University College London (UCL), UK Mark: viva passed with no corrections. Award date: 28/03/2016. Thesis: " <i>Microstructural imaging of the human spinal cord with advanced diffusion MRI</i> "
Master's Degree in Bioengineering	2010-2012	University of Genoa, Italy Mark: 110 out of 110 cum laude & Dignità di Stampa (Examination Panel award). Dissertation: " <i>A study on a bidirectional brain-machine interface inspired by the corticospinal control of movement</i> ".
Bachelor's Degree in Biomedical Engineering	2006-2009	University of Cagliari, Italy Mark: 110 out of 110 cum laude. Dissertation: " <i>Real time wavelet denoising on a DSP of neural signals coming from the peripheral nervous system</i> ".

WORK EXPERIENCE

Vall d'Hebron Institute of Oncology	Jan.2025-now	Senior investigator, VHIO — SPAIN Affiliations: Radiomics Group, Biomedical Research Department. Role: supervisor of research line in diffusion MRI. Design of MRI studies.
Vall d'Hebron Institute of Oncology	Oct.2020-Dec.24	Senior post-doc, VHIO — BARCELONA, SPAIN Affiliations: Radiomics Group, Clinical Research Department. Role: development of microstructural MRI techniques in cancer.
University College London	Feb.2016-Sept.20	Research Associate, UCL — LONDON, UK Affiliations: Institute of Neurology; Centre for Medical Image Computing. Responsibilities: quantitative MRI development (spinal cord, brain, prostate).
Queen Mary Univ. of London	May-June 2012	Research Assistant, QMUL — LONDON, UK Affiliation: School of Electronic Engineering and Computer Science (EECS). Responsibilities: image segmentation (Computed Tomography Angiography).

RESEARCH VISITS

Feb.-March 2024 Champalimaud Foundation, Portugal
 Visiting
 Researcher **Affiliation:** Preclinical MRI lab, Champalimaud Foundation, Lisbon, Portugal.
Training: preclinical MRI and multi-echo gradient echo imaging.

Sept.-Nov. 2017 New York University (NYU), USA
 Visiting
 Researcher **Affiliation:** Radiology, Langone Medical Center, New York City (USA).
Training: advanced denoising techniques for diffusion MRI.

IMPACT STATS ON THE 05/10/2026

h-index Google Scholar h-index: 27. Web of Science h-index: 21.

Citations Total Google Scholar citations: 3015. Web of Science citations: 1772.

FUNDING: PROJECT GRANTS

EU4Health 2026 **EU4Health project grant**, call EU4H-2026-SANTE-PJ-o8, European Commission. “Scale-up biotech AI innovation by leveraging responsible data use under the European Health Data Space” (SCAILED). **Award:** 92,362.4 €. **Role:** VHIO official co-investigator. Principal investigator and primary coordinator: Gu Wei, Luxembourg National Data Service (LNDS). **Consortium:** 47 institutes/centres, with total budget of 14,384,759.99 €. **Duration:** 01/10/2026-30/09/2029.

CaixaResearch 2025 **CaixaResearch Institute Innovation Hub 2025**. CaixaResearch Institute, Barcelona (Spain). “Computational Magnetic Resonance for metastatic prostate cancer”. **Award:** 30,000 €. **Role:** principal investigator. **Duration:** 01/01/2026-31/12/2026. Competitive call open to researchers from Innovation Hub institutions (VHIO; IrsiCaixa; ISGlobal; BarcelonaBeta).

AEI Generación de Conocimiento/ Knowledge Generation 2024 **2024 Knowledge Generation Projects (Proyectos de Generación de Conocimiento)**, National Research Agency (Agencia Estatal de Investigación (AEI)), Ministry of Science, Innovation and Universities, Spain. “Magnetic Resonance Imaging (MRI) foundational artificial intelligence for non-invasive, histologically-meaningful cancer characterisation (MRI-Found-Histo)”. **Award:** 118,750 €. **Role:** principal investigator. **Duration:** 01/09/2025-31/08/2028. **Code:** PID2024-158670OA-I00.

UCL pump-priming award 2017 **Departmental seed grant**. Centre for Medical Image Computing Pump-priming Award at University College London (UCL). “Enabling multi-site high precision spinal cord MRI”. **Award:** GBP 23,900. **Role:** principal investigator. **Duration:** 01/07/2017-30/06/2018.

FUNDING: FELLOWSHIPS AND STUDENTSHIPS

AEI “Ramón y Cajal” Fellowship 2025 **2025 Ramón y Cajal tenure-track fellowship**, National Research Agency (Agencia Estatal de Investigación (AEI)), Ministry of Science, Innovation and Universities, Spain. “Computational Magnetic Resonance biophysics for enhanced neurological and oncological imaging”. **Award:** €259,810. **Role:** principal investigator and fellow. **Duration:** 2027-2031 (awarded on 07/09/2026, agreement with Universitat Politècnica de Catalunya submitted to AEI on 28/09/2026). **Code:** RYC2025-054665-I.

“la Caixa” Junior Leader Fellowship 2022 **2022 Junior Leader Retaining post-doctoral fellowship**, “la Caixa” Foundation, Spain. “New-generation oncological MRI (New-OncoMRI): development, validation and application”. **Award:** €294,900. **Role:** principal investigator and fellow. **Duration:** 30/09/2022-29/09/2025. **Code:** ID 100010434, fellowship number LCF/BQ/PR22/11920010.

AGAUR Beatriu
de Pinós
Fellowship 2020

2020 Beatriu de Pinós post-doctoral fellowship, AGAUR, Secretary of Universities and Research (Govt of Catalonia, Spain). “*Advancing Magnetic Resonance Imaging against liver cancer*”. **Award:** €144,300. **Role:** principal investigator and fellow. **Duration:** 01/01/2022-31/12/2024, renounced on 29/09/2022 due to incompatibility with the “la Caixa” Junior Leader fellowship. **Code:** 2020 BP 00117.

UCL Grand
Challenge PhD
studentship 2012

UCL School of Life and Medical Science *Grand Challenge PhD Studentship*, awarded to work on the project “*Axonal density as MR imaging biomarker: from bench to bedside*”. **Award:** 59,000 GBP. **Description:** PhD studentship covering a 3-year student stipend, tuition fees at UK/EU rate, and research costs. **Duration:** 09/2012-12/2015.

POSITIONS: DECLINED COMPETITIVE AWARDS

ISCIH “Miguel
Servet”
tenure-track
Fellowship 2026

2026 Miguel Servet tenure-track fellowship, Carlos III Health Institute (ISCIH, Instituto de Salud Carlos III, Ministry of Health and Ministry of Science, Innovation and Universities, Spain). “*Bridging histology and medical imaging with biophysical digital twins to enhance precision oncology*”. **Award:** €308,600. **Role:** principal investigator and fellow. **Code:** CP26/00050. Provisional award 15/07/2026, to be declined due to incompatibility with RyC fellowship.

2025 UAB “Serra
Hünter”
tenure-track
Lecturership

Serra Hünter tenure-track Lecturership/Assistant Professorship (*professor ajudant doctor*) funded by the Generalitat de Catalunya. Department of Computer Sciences, Universitat Autònoma de Barcelona, Cerdanyola/Barcelona, Spain. **Duration:** 2026-2032 (awarded 30/06/2026, declined due to incompatibility with RyC fellowship). **Code:** UAB-LE-251-010.

SCIENTIFIC ARTICLES AND REVIEWS — SENIOR AUTHORSHIP

Magn Reson Med
2026

“*Simulation-informed evaluation of microvascular parameter mapping for diffusion MR imaging of solid tumours*”. Voronova AK et al, Grussu F. *Magnetic Resonance in Medicine* (2026), 96(1): 387-402, doi: [10.1002/mrm.70318](https://doi.org/10.1002/mrm.70318). Senior (last) author, corresponding author. **IF:** 3.3 (Quartile Q2) in the year of publication.

Comm Bio 2025

“*Histology-informed microstructural diffusion simulations for MRI cancer characterisation — the Histo-microSim framework*”. Grigoriou A et al, Grussu F. *Communications Biology* (2025), 8: 1695, doi: [10.1038/s42003-025-09096-3](https://doi.org/10.1038/s42003-025-09096-3). Senior (last) author, corresponding author. **IF:** 5.1 (Decile D1) in the year of publication.

MedIA 2025

“*SpinFlowSim: a blood flow simulation framework for histology-informed diffusion MRI microvasculature mapping in cancer*”. Voronova AK et al, Grussu F. *Medical Image Analysis* (2025), 102: 103531, doi: [10.1016/j.media.2025.103531](https://doi.org/10.1016/j.media.2025.103531). Senior (last) author, corresponding author. **IF:** 11.8 (Decile D1) in the year of publication.

JMRI 2025

“*Enhancing tumor microstructural quantification with machine learning and diffusion-relaxation MRI*”. Macarro C et al, Grussu F. *Journal of Magnetic Resonance Imaging* (2025), 61(2): 1018-1021, doi: [10.1002/jmri.29484](https://doi.org/10.1002/jmri.29484). Senior (last) author, corresponding author. **IF:** 3.5 (Quartile Q1) in the year of publication.

JMRI 2024

Review: “*Advanced diffusion-weighted MRI for cancer microstructure assessment in body imaging, and its relationship with histology*”. Fokkinga E, Hernandez-Tamames JA, Ianus A, Nilsson M, Tax CMW, Perez-Lopez R, Grussu F. *Journal of Magnetic Resonance Imaging* (2024), 60(4): 1278-1304, doi: [10.1002/jmri.29144](https://doi.org/10.1002/jmri.29144). Senior (last) author, corresponding author. **IF:** 3.3 (Quartile Q1) in the year of publication.

Front Neurol 2021

“*Comparison of neurite orientation dispersion and density imaging and two-compartment spherical mean technique parameter maps in multiple sclerosis*”. Johnson D et al, Grussu F. *Frontiers in Neurology* (2021), 12: 662855, doi: [10.3389/fnol.2021.662855](https://doi.org/10.3389/fnol.2021.662855).

10.3389/fneur.2021.662855. Senior (last) author. IF: 4.086 (Quartile Q2) in the year of publication.

SCIENTIFIC ARTICLES AND REVIEWS — (CO)-FIRST AUTHORSHIP

- J Immunother Cancer* 2025 “Integrating C-reactive protein flare and early MRI dynamics for enhanced prediction of immunotherapy response”. Navarro D*, Grussu F* et al. *Journal for ImmunoTherapy of Cancer* (2025), 13: e012143, doi: 10.1136/jitc-2025-012143. *: DN and FG are joint first authors (equal contribution). IF: 10.6 (Decile D1) in the year of publication.
- Comm Med* 2025 “Clinically feasible liver tumour cell size measurement through histology-informed in vivo diffusion MRI”. Grussu F et al. *Communications Medicine* (2025), 5: 535, doi: 10.1038/s43856-025-01246-2. Corresponding author. IF: 6.3 (Quartile Q1) in the year of publication.
- Cell Rep Med* 2024 “An accessible deep learning tool for voxel-wise classification of brain malignancies from perfusion MRI”. Garcia-Ruiz A*, Pons-Escoda A*, Grussu F* et al. *Cell Reports Medicine* (2024), 5(3): 101464, doi: 10.1016/j.xcrm.2024.101464. *: AGR, APE and FG are joint first authors (equal contribution). IF: 11.7 (Decile D1) in the year of publication.
- Magn Reson Med* 2022 “Diffusion MRI signal cumulants and hepatocyte microstructure at fixed diffusion time: Insights from simulations, 9.4T imaging, and histology”. Grussu F et al. *Magnetic Resonance in Medicine* (2022), 88(1): 365-379, doi: 10.1002/mrm.29174. Corresponding author. IF: 3.737 (Quartile Q2) in the year of publication.
- Front Phys* 2021 “Feasibility of data-driven, model-free quantitative MRI protocol design: application to brain and prostate diffusion-relaxation imaging”. Grussu F et al. *Frontiers in Physics* (2021), 9: 752208, doi: 10.3389/fphy.2021.752208. Corresponding author. IF: 3.560 (Quartile Q2) in the year of publication.
- Sem Ultrasound CT MRI* 2021 **Review:** “Diffusion-weighted imaging: recent advances and applications”. Martinez-Heras E*, Grussu F*, et al. *Seminars in Ultrasound, CT and MRI* (2021), 42(5): 490-506, doi: 10.1053/j.sult.2021.07.006. *: EMH and FG are joint first authors (equal contribution). IF: 1.600 (Quartile Q3) in the year of publication.
- NeuroImage* 2020 “Multi-parametric quantitative in vivo spinal cord MRI with unified signal readout and image denoising”. Grussu F et al. *NeuroImage* (2020), 217: 116884, doi: 10.1016/j.neuroimage.2020.116884. Corresponding author. IF: 5.902 (Decile D1) in the year of publication.
- Magn Reson Med* 2019 “Relevance of time-dependence for clinically viable diffusion imaging of the spinal cord”. Grussu F et al. *Magnetic Resonance in Medicine* (2019), 81(2): 1247-1264, doi: 10.1002/mrm.27463. Corresponding author. IF: 3.858 (Quartile Q1) in the year of publication.
- Ann Clin Transl Neurol* 2017 “Neurite dispersion: a new marker of multiple sclerosis spinal cord pathology?”. Grussu F, Schneider T et al. *Annals of Clinical and Translational Neurology* (2017), 4(9):663-679, doi: 10.1002/acn3.445. FG and TS are joint first authors (equal contribution). IF: 3.901 (Quartile Q1) in the year of publication. Paper featured in *Nature Reviews Neurology* “Research Highlights” (Patel M, *Nat Rev Neur* (2017), 13(10): 578, doi: 10.1038/nrneurol.2017.127).
- J Neurosci Meth* 2016 “A framework for optimal whole-sample histological quantification of neurite orientation dispersion in the human spinal cord”. Grussu F et al. *Journal of Neuroscience Methods* (2016), 273: 20-32, doi: 10.1016/j.jneumeth.2016.08.002. Corresponding author. IF: 2.053 (Quartile Q3) in the year of publication.
- NeuroImage* 2015 “Neurite orientation dispersion and density imaging of the healthy cervical spinal cord in vivo”. Grussu F et al. *NeuroImage* (2015), 111: 590-601, doi: 10.1016/j.neuroimage.2015.01.045. Corresponding author. IF: 6.357 (Decile D1) in the year of publication.

- Eur Radiol* 2026 **Review:** “ESR Essentials: diffusion-weighted MRI—practice recommendations by the European Society for Magnetic Resonance in Medicine and Biology”. Palombo M, Bodini B, Grussu F et al. *European Radiology* (2026), 36: 2198-2208, doi: [10.1007/s00330-025-12033-x](https://doi.org/10.1007/s00330-025-12033-x). IF: 4.7 (Quartile Q1) in the year of publication.
- JMRI* 2026 “The sense of smell (SoS) atlas: its creation and first application to investigate COVID-19 related anosmia with a comprehensive quantitative MRI protocol”. Gaviraghi M, ..., Grussu F et al. *Journal of Magnetic Resonance Imaging* (2026), 63(2): 574-593, doi: [10.1002/jmri.70128](https://doi.org/10.1002/jmri.70128). IF: 3.5 (Quartile Q1) in the year of publication.
- Clin Cancer Res* 2026 “Broad utility of ultrasensitive analysis of ctDNA dynamics across solid tumors treated with immunotherapy”. Garralda E, ..., Grussu F et al. *Clinical Cancer Research* (2026), 32(2): 333-349, doi: [10.1158/1078-0432.CCR-25-2312](https://doi.org/10.1158/1078-0432.CCR-25-2312). IF: 10.2 (Decile D1) in the year of publication.
- Magn Res Med* 2025c **Review:** “Considerations and recommendations from the ISMRM Diffusion Study Group for preclinical diffusion MRI: Part 3—Ex vivo imaging: Data processing, comparisons with microscopy, and tractography”. Schilling KG, Howard AFD, Grussu F et al. *Magnetic Resonance in Medicine* (2025), 93(6): 2561-2582, doi: [10.1002/mrm.30424](https://doi.org/10.1002/mrm.30424). IF: 3.0 (Quartile Q2) in the year of publication.
- Magn Res Med* 2025b **Review:** “Considerations and recommendations from the ISMRM diffusion study group for preclinical diffusion MRI: Part 2—Ex vivo imaging: Added value and acquisition”. Schilling KG, Grussu F et al. *Magnetic Resonance in Medicine* (2025), 93(6): 2535-2560, doi: [10.1002/mrm.30435](https://doi.org/10.1002/mrm.30435). IF: 3.0 (Quartile Q2) in the year of publication.
- Magn Res Med* 2025a **Review:** “Considerations and recommendations from the ISMRM diffusion study group for preclinical diffusion MRI: Part 1: In vivo small-animal imaging”. Jelescu IO, Grussu F et al. *Magnetic Resonance in Medicine* (2025), 93(6): 2507-2534, doi: [10.1002/mrm.30429](https://doi.org/10.1002/mrm.30429). IF: 3.0 (Quartile Q2) in the year of publication.
- Imag Neurosci* 2025 “Body size and intracranial volume interact with the structure of the central nervous system: A multi-center in vivo neuroimaging study”. Labounek R, ... Grussu F et al. *Imaging Neuroscience* (2025), 3: 00559, doi: [10.1162/imag.a.00559](https://doi.org/10.1162/imag.a.00559). No IF available in the year of publication.
- Sci Rep* 2025 “Evaluation of magnetic resonance spectroscopy total sodium concentration measures, and associations with microstructure and physical impairment in cervical myelopathy”. Solanky B, ... Grussu F et al. *Scientific Reports* (2025), 15: 7014, doi: [10.1038/s41598-025-91658-w](https://doi.org/10.1038/s41598-025-91658-w). IF: 3.9 (Quartile Q1) in the year of publication.
- J Immunother Cancer* 2025 “Radiomics signature for dynamic monitoring of tumor inflamed microenvironment and immunotherapy response prediction”. Bernatowicz K, ... Grussu F et al. *Journal for ImmunoTherapy of Cancer* (2025), 13: e009140, doi: [10.1136/jitc-2024-009140](https://doi.org/10.1136/jitc-2024-009140). IF: 10.6 (decile D1) in the year of publication.
- NeuroImage Rep* 2024 “Investigating the relationship between thalamic iron concentration and disease severity in secondary progressive multiple sclerosis using quantitative susceptibility mapping: Cross-sectional analysis from the MS-STAT2 randomised controlled trial”. Williams T, ..., Grussu F, ..., Chataway J; On behalf of The UCL MS-STAT2 investigators. *NeuroImage: Reports* (2024), 4(3): 100216, doi: [10.1016/j.ynirp.2024.100216](https://doi.org/10.1016/j.ynirp.2024.100216). No IF available in the year of publication.
- Eur Urol* 2024 “Whole-body magnetic resonance imaging as a treatment response biomarker in castration-resistant prostate cancer with bone metastases: the iPROMET clinical trial”. Garcia-Ruiz A, ... Grussu F et al. *European Urology* (2024), 86(3): 272-274, doi: [10.1016/j.eururo.2024.02.016](https://doi.org/10.1016/j.eururo.2024.02.016). IF: 25.3 (Decile D1) in the year of publication.
- Mult Scler* 2024 “What contributes to disability in progressive MS? A brain and cervical cord-matched quantitative MRI study”. Tur C, ..., Grussu F et al. *Multiple Sclerosis Journal*

(2024), e-pub ahead of print, doi: [10.1177/13524585241229969](https://doi.org/10.1177/13524585241229969). IF: 4.8 (Quartile Q1) in the year of publication.

- Radiology: AI*
2024
"Identification of precise 3D CT radiomics for habitat computation by machine learning in cancer". Prior O, ..., Grussu F et al. *Radiology: Artificial Intelligence* (2024), 6(2): e230118, doi: [10.1148/ryai.230118](https://doi.org/10.1148/ryai.230118). IF: 8.1 (Quartile Q1) in the year of publication.
- Movement Disord*
2023
"Multimodal analysis of the visual pathways in Friedreich's Ataxia reveals novel biomarkers". Thomas-Black G, ... Grussu F et al. *Movement Disorders* (2023), 38(6): 959-969, doi: [10.1002/mds.29277](https://doi.org/10.1002/mds.29277). IF: 8.6 (Decile D1) in the year of publication.
- Sci Rep* 2023
"Feasibility of in vivo multi-parametric quantitative magnetic resonance imaging of the healthy sciatic nerve with a unified signal readout protocol". Boonsuth R, Battiston M, Grussu F et al. *Scientific Reports* (2023), 13: 6565, doi: [10.1038/s41598-023-33618-w](https://doi.org/10.1038/s41598-023-33618-w). IF: 4.6 (Quartile Q2) in the year of publication.
- Front Neuroinform*
2023
"Patterns of inflammation, microstructural alterations, and sodium accumulation define multiple sclerosis subtypes after 15 years from onset". Ricciardi A, Grussu F et al. *Frontiers in Neuroinformatics* (2023), 17: 1060511, doi: [10.3389/fninf.2023.1060511](https://doi.org/10.3389/fninf.2023.1060511). IF: 3.5 (Quartile Q2) in the year of publication.
- Neurology* 2023
"Differentiating Multiple Sclerosis from AQP4-Neuromyelitis Optica Spectrum Disorder and MOG-antibody disease with imaging". Cortese R, ... Grussu F et al. *Neurology* (2023), 100(3): e308-e323, doi: [10.1212/WNL.0000000000201465](https://doi.org/10.1212/WNL.0000000000201465). IF: 10.1 (Decile D1) in the year of publication.
- Am J Neuroradiol*
2022
"Diffuse large B-cell Epstein-Barr virus-positive primary CNS lymphoma in non-AIDS patients: high diagnostic accuracy of DSC perfusion metrics". Pons-Escoda A, ... Grussu F et al. *American Journal of Neuroradiology* (2022), 43(11): 1567-1574, doi: [10.3174/ajnr.A7668](https://doi.org/10.3174/ajnr.A7668). IF: 4.966 (Quartile Q2) in the year of publication.
- Magn Reson Med*
2022c
"Multi-echo quantitative susceptibility mapping: how to combine echoes for accuracy and precision at 3 Tesla". Biondetti E, ..., Grussu F et al. *Magnetic Resonance in Medicine* (2022), 88(5): 2101-2116, doi: [10.1002/mrm.29365](https://doi.org/10.1002/mrm.29365). IF: 3.737 (Quartile Q2) in the year of publication.
- Magn Reson Med*
2022b
"SENSE EPI reconstruction with 2D phase error correction and channel-wise noise removal". Powell E, ..., Grussu F et al. *Magnetic Resonance in Medicine* (2022), 88(5): 2157-2166, doi: [10.1002/mrm.29349](https://doi.org/10.1002/mrm.29349). IF: 3.737 (Quartile Q2) in the year of publication.
- Magn Reson Med*
2022a
"Comparison of multicenter MRI protocols for visualizing the spinal cord gray matter". Cohen-Adad J, ..., Grussu F et al. *Magnetic Resonance in Medicine* (2022), 88(2): 849-859, doi: [10.1002/mrm.29249](https://doi.org/10.1002/mrm.29249). IF: 3.737 (Quartile Q2) in the year of publication.
- Eur Radiol* 2022
"Voxel-level analysis of normalized DSC-PWI time-intensity curves: a potential generalizable approach and its proof of concept in discriminating glioblastoma and metastasis". Pons-Escoda A, ..., Grussu F et al. *European Radiology* (2022), 32: 3705-3715, doi: [10.1007/s00330-021-08498-1](https://doi.org/10.1007/s00330-021-08498-1). IF: 7.034 (Quartile Q1) in the year of publication.
- BMJ Open* 2022
"Histo-MRI map study protocol: a prospective cohort study mapping MRI to histology for biomarker validation and prediction of prostate cancer". Singh S, ..., Grussu F et al. *BMJ Open* (2022), 12: e059847, doi: [10.1136/bmjopen-2021-059847](https://doi.org/10.1136/bmjopen-2021-059847). IF: 3.007 (Quartile Q2) in the year of publication.
- NeuroImage Clin*
2022
"Spatial patterns of brain lesions assessed through covariance estimations of lesional voxels in multiple sclerosis: the SPACE-MS technique". Tur C, Grussu F et al. *NeuroImage: Clinical* (2022), 33: 102904, doi: [10.1016/j.nicl.2021.102904](https://doi.org/10.1016/j.nicl.2021.102904). IF: 4.891 (Quartile Q2) in the year of publication.
- Front Neurol* 2021
"Assessing lumbar plexus and sciatic nerve damage in relapsing-remitting multiple

- sclerosis using magnetisation transfer ratio". Boonsuth R, ..., Grussu F et al. Frontiers in Neurology (2021), 12: 763143, doi: 10.3389/fneur.2021.763143. IF: 4.003 (Quartile Q2) in the year of publication.*
- Sci Rep 2021* "Robust imaging habitat computation using voxel-wise radiomics features". Bernatowicz K, Grussu F et al. Scientific Reports (2021), 11: 20133, doi: 10.1038/s41598-021-99701-2. IF: 4.380 (Quartile Q1) in the year of publication.
- Sci Data 2021* "Open-access quantitative MRI data of the spinal cord and reproducibility across participants, sites and manufacturers". Cohen-Adad J, ..., Grussu F et al. Scientific Data (2021), 8: 219, doi: 10.1038/s41597-021-00941-8. IF: 6.444 (Quartile Q1) in the year of publication.
- Nat Protoc 2021* "Generic acquisition protocol for quantitative MRI of the spinal cord". Cohen-Adad J, ..., Grussu F et al. Nature Protocols (2021), 16: 4611–4632, doi: 10.1038/s41596-021-00588-0. IF: 13.491 (decile D1) in the year of publication.
- Brain 2021* "Brain microstructural and metabolic alterations detected in vivo at the onset of the first demyelinating event". Collorone S, ..., Grussu F et al. Brain (2021), 144: 1409–1421, doi: 10.1093/brain/awab043. IF: 13.501 (Decile D1) in the year of publication.
- NeuroImage 2021* "Uncertainty modelling in deep learning for safer neuroimage enhancement: demonstration in diffusion MRI". Tanno R, ... Grussu F et al. NeuroImage (2021), 225: 117366, doi: 10.1016/j.neuroimage.2020.117366. IF: 6.556 (Decile D1) in the year of publication.
- Mult Scler 2020b* "Reduced neurite density in the brain and cervical spinal cord in relapsing–remitting multiple sclerosis: A NODDI study". Collorone S, Cowley N, Grussu F et al. Multiple Sclerosis Journal (2020), 26(13): 1647–1657, doi: 10.1177/1352458519885107. IF: 5.412 (Quartile Q1) in the year of publication.
- Mult Scler 2020a* "A multi-shell multi-tissue diffusion study of brain connectivity in early multiple sclerosis". Tur C, Grussu F et al. Multiple Sclerosis Journal (2020), 26(7): 774–785, doi: 10.1177/1352458519845105. IF: 5.412 (Quartile Q1) in the year of publication.
- NeuroImage 2020b* "Cross-scanner and cross-protocol multi-shell diffusion MRI data harmonization: Algorithms and results". Ning L, Bonet-Carne E, Grussu F et al. NeuroImage (2020), 221: 117128, doi: 10.1016/j.neuroimage.2020.117128. IF: 5.902 (Decile D1) in the year of publication.
- NeuroImage 2020a* "Generalised boundary shift integral for longitudinal assessment of spinal cord atrophy". Prados F, ..., Grussu F et al. NeuroImage (2020), 209: 116489, doi: 10.1016/j.neuroimage.2019.116489. IF: 5.902 (Decile D1) in the year of publication.
- Magn Reson Med 2019* "Fast bound pool fraction mapping via steady-state magnetization transfer saturation using single-shot EPI". Battiston M, ..., Grussu F et al. Magnetic Resonance in Medicine (2019), 82: 1025–1040, doi: 10.1002/mrm.27792. IF: 3.858 (Quartile Q1) in the year of publication.
- NeuroImage 2019* "Cross-scanner and cross-protocol diffusion MRI data harmonisation: a benchmark database and evaluation of algorithms". Tax CMW, Grussu F et al. NeuroImage (2019), 195: 285–299, doi: 10.1016/j.neuroimage.2019.01.077. IF: 5.812 (Decile D1) in the year of publication.
- Sci Rep 2018* "Structural cortical network reorganization associated with early conversion to multiple sclerosis". Tur C, ..., Grussu F et al. Scientific Reports (2018), 8: 10715, doi: 10.1038/s41598-018-29017-1. IF: 4.122 (Quartile Q1) in the year of publication.
- Magn Reson Med 2018b* "An optimized framework for quantitative magnetization transfer imaging of the cervical spinal cord in vivo". Battiston M, Grussu F et al. Magnetic Resonance in Medicine (2018) 79(5): 2576–2588, doi: 10.1002/mrm.26909. IF: 3.924 (Quartile Q1) in the year of publication.

- Magn Reson Med 2018a *"Fast and reproducible in vivo T₁ mapping of the human cervical spinal cord"*. Battiston M, ..., Grussu F et al. Magnetic Resonance in Medicine (2018), 79(4): 2142-2148, doi: [10.1002/mrm.26852](https://doi.org/10.1002/mrm.26852). IF: 3.924 (Quartile Q1) in the year of publication.
- NeuroImage 2017 *"Spinal cord grey matter segmentation challenge"*. Prados F, ..., Grussu F et al. NeuroImage (2017), 152:312-329, doi: [10.1016/j.neuroimage.2017.03.010](https://doi.org/10.1016/j.neuroimage.2017.03.010). IF: 5.835 (Decile D1) in the year of publication.
- PlosOne 2016 *"Reduced field-of-view diffusion-weighted imaging of the lumbosacral enlargement: a pilot in vivo study of the healthy spinal cord at 3T"*. Yiannakas MC, Grussu F et al. PlosOne (2016), 11(10): e0164890, doi: [10.1371/journal.pone.0164890](https://doi.org/10.1371/journal.pone.0164890). IF: 3.057 (Quartile Q1) in the year of publication.

CONFERENCE PAPERS: FIRST AUTHORSHIP

- CDMRI 2020 *"Deep learning model fitting for diffusion-relaxometry: a comparative study"*. Grussu F, Battiston M, Palombo M, Schneider T, Wheeler-Kingshott CAM, Alexander DC. Proc of 2020 MICCAI Workshop on Computational Diffusion MRI, 2021, 159-172, doi: [10.1007/978-3-030-73018-5_13](https://doi.org/10.1007/978-3-030-73018-5_13). Corresponding author.

CONFERENCE PAPERS: CO-AUTHORSHIP

- MICCAI 2022 *"Progressive subsampling for oversampled data - application to quantitative MRI"*. Blumberg SB, ..., Grussu F et al. Proc of Medical Image Computing and Computing Assisted Intervention (MICCAI) 2022, Lecture Notes in Computer Science, 13436: 421-431, doi: [10.1007/978-3-031-16446-0_40](https://doi.org/10.1007/978-3-031-16446-0_40).
- CDMRI 2019 *"Acquiring and predicting multidimensional diffusion (MUDI) data: an open challenge"*. Pizzolato M, ..., Grussu F et al. Proc of 2019 MICCAI Workshop on Computational Diffusion MRI, 2020, 195-208, doi: [10.1007/978-3-030-52893-5_17](https://doi.org/10.1007/978-3-030-52893-5_17).
- CDMRI 2018b *"Multi-shell diffusion MRI harmonisation and enhancement challenge (MUSHAC): progress and results"*. Ning L, ..., Grussu F et al. Proc of 2018 MICCAI Workshop on Computational Diffusion MRI, 2019, 217-224, doi: [10.1007/978-3-030-05831-9_18](https://doi.org/10.1007/978-3-030-05831-9_18).
- CDMRI 2018a *"Spatial characterisation of fibre response functions for spherical deconvolution in multiple sclerosis"*. Tur C, Grussu F et al. Proc of 2018 MICCAI Workshop on Computational Diffusion MRI, 2019, 265-279, doi: [10.1007/978-3-030-05831-9_21](https://doi.org/10.1007/978-3-030-05831-9_21).
- MICCAI 2016 *"Bayesian image quality transfer"*. Tanno R, ..., Grussu F et al. Proc of Medical Image Computing and Computing Assisted Intervention (MICCAI) 2016, Lecture Notes in Computer Science, 9901: 265-273, doi: [10.1007/978-3-319-46723-8_31](https://doi.org/10.1007/978-3-319-46723-8_31).

BOOK CHAPTERS

- CRC Press 2018 Chapter 8: *"D — the diffusion of water (DTI)"*. Grussu F and Wheeler-Kingshott CAM. *"Quantitative MRI of the brain"* (2nd edition, 2018), Cercignani M, Dowell N and Tofts P editors. ISBN 978-1-138-03285-9, doi: [10.1201/b21837](https://doi.org/10.1201/b21837).

BOOK EDITING

- CDMRI 2018 *"Computational Diffusion MRI"*. Bonet-Carne E, Grussu F, Ning L, Seppehrband F and Tax C editors. Proc. of 2018 MICCAI Workshop on *"Computational Diffusion MRI"*, Granada, Spain, 20/09/2018. ISBN: 978-3-030-05830-2, doi: [10.1007/978-3-030-05831-9](https://doi.org/10.1007/978-3-030-05831-9).

CDMRI 2017 “Computational Diffusion MRI”. Kaden E, Grussu F, Ning L, Tax C and Veraart J editors. Proc. of 2017 MICCAI Workshop on “Computational Diffusion MRI”, Quebec City, Canada, 10/09/2017. ISBN: 978-3-319-73839-0, doi: [10.1007/978-3-319-73839-0](https://doi.org/10.1007/978-3-319-73839-0).

PRIZES AND AWARDS

- 2023 **Best oral paper award**, 2023 annual meeting of the **Iberian Chapter** of the International Society for Magnetic Resonance in Medicine (ISMRM), Trainee competition for abstract Grussu F et al, Proc of ISMRM Iberian Chapt. 2023.
- 2021 **3rd prize**, 2021 ISMRM **MR of Cancer Study Group**, Trainee competition for abstract Grussu F et al, Proc of ISMRM 2021, p.0699.
- 2021-2015 **Magna cum Laude** awards at the ISMRM annual meeting in 2021, 2020, 2018 and 2015.
- 2020 **2nd prize** (shared), 2020 ISMRM British and Irish Chapter “**Mansfield Research Innovation Award**” for abstract Grussu F et al, Proc of ISMRM 2020.
- 2019 **1st prize** in the “Multi-dimensional Diffusion Imaging” (**MUDI**) challenge at 2019 CDMRI MICCAI Workshop (Shenzhen, China, 17/10/2019) (**Team**: Grussu F, Blumberg SB, Ianus A, Mertzaniadou T, Alexander DC; **Method**: SARDU-Net).

AWARDS IN SUPERVISORY ROLES

- 2024 **Best application (poster)** by the Diffusion Study Group of the International Society for Magnetic Resonance in Medicine (ISMRM) to student Anna Voronova, for abstract (Voronova A et al, and Grussu F, p.0124, ISMRM 2024).
- 2024 **Summa cum Laude** award to student Anna Voronova, for abstract (Voronova A et al, and Grussu F, p.0124, ISMRM 2024).
- 2024 **Magna cum Laude** award to student Athanasios Grigoriou, for abstract (Grigoriou A et al, and Grussu F, p.0699, ISMRM 2024).

ELECTED INTERNATIONAL POSITIONS

- ISMRM 2018-2020 **Elected trainee (student) representative** for the **White Matter Study Group** of the International Society for Magnetic Resonance in Medicine (ISMRM).

INVITED ORAL COMMUNICATIONS

- ISMRM Iberian Chapter 2026 “From histology to radiological maps: a journey through MRI signal modelling and beyond”. International Society for Magnetic Resonance in Medicine (ISMRM), 2026 annual meeting of the Iberian Chapter. “**Keynote lecture**”. University of Coimbra, Coimbra (Portugal), 02/07/2026.
- BSC 2025 “Histology-informed computer simulations enhance Magnetic Resonance Imaging (MRI) virtual biopsies in cancer”. “**Severo Ochoa Research Seminar**”, Barcelona Supercomputing Center (BSC), Barcelona, Spain, 13/11/2025.
- ODELIA Summer School 2025 “An overview of Magnetic Resonance Imaging (MRI): from basic principles to cutting-edge applications”. ODELIA consortium on open source swarm learning to empower medical AI. “**2025 ODELIA Summer School**”, Vall d’Hebron Institute of Oncology (VHIO), Barcelona, Spain, 18/09/2025.
- ISMRM Iberian Chapter 2025 “Quantitative imaging and AI Automation”. International Society for Magnetic Resonance in Medicine (ISMRM), 2025 annual meeting of the Iberian Chapter. “**2025 ISMRM Iberian Chapter pre-conference Summer School**”, Institut de Bioenginyeria de Catalunya (IBEC), Barcelona (Spain), 02/07/2025.

ISMRM Workshop 2025	<i>"Challenges in body diffusion and how to overcome them"</i> . International Society for Magnetic Resonance in Medicine (ISMRM) Workshop on: <i>"40 years of Diffusion: past, present and future perspectives"</i> , Kyoto University, Kyoto (Japan), 18/02/2025.
EPFL 2024	<i>"Diffusion MR signal modelling for oncological body imaging"</i> . <i>Visitors Talk</i> , Centre for Biomedical Imaging (CIBM), École Polytechnique Fédérale de Lausanne (EPFL), Lausanne (Switzerland), 09/10/2024.
ESMRMB workshop 2024	<i>"Modelling body microstructure"</i> . ESMRMB 2024 pre-congress workshop <i>"Leaps in Microstructure Imaging: Exploring New Horizons"</i> , Barcelona (Spain), 02/10/2024.
DiPy workshop 2024	<i>"Advanced body diffusion MRI for oncological applications"</i> . 2024 DiPy workshop, online, 14/03/2024.
ESMRMB workshop 2023	<i>"Histology-informed body diffusion MRI in oncological applications"</i> . ESMRMB 2023 pre-congress workshop <i>"Frontiers in preclinical MRI"</i> , Basel (Switzerland), 04/10/2023.
ESMRMB lectures 2023	<i>"Diffusion MRI in the body"</i> . ESMRMB Lectures on MR, <i>"Introduction to diffusion-weighted MR imaging and spectroscopy"</i> , Cardiff (UK), 07/09/2023.
DSG ISMRM 2023	<i>"Emerging models in oncology"</i> . <i>"Virtual biopsies by diffusion MRI: are we there yet in oncology?"</i> , Virtual meeting of the International Society for Magnetic Resonance in Medicine (ISMRM) Diffusion Study Group (DSG). Online, 19/04/2023.
UCL 2022b	<i>"Data-driven, model-free, deep learning approach for quantitative MRI protocol design"</i> . <i>"Microstructure Imaging Meets Machine Learning"</i> (MIML) workshop, University College London, London (UK), 13/05/2022.
ISMRM 2022	<i>"Modeling diffusion in cancer and body"</i> . Educational session: <i>"Diffusion"</i> , 2022 annual meeting of the International Society for Magnetic Resonance in Medicine (ISMRM), London (UK), 07/05/2022.
University of Verona 2019	<i>"Diffusion MRI data harmonisation"</i> . 2019 School on Brain Connectomics, University of Verona (Italy), 24/09/2019.
ISMRM Italian Chapter 2018	<i>"Axonal dispersion from diffusion MRI: a new marker of microstructural damage"</i> . Italian Association for Magnetic Res. in Medicine, Padua (Italy), 10/05/2018.

MODERATION OF SCIENTIFIC SESSIONS

ECR 2026	Moderator of European Society of Radiology (ESR) Research Committee Session 10 <i>"Real-world applications of imaging biomarkers: from hypothesis to clinical practice"</i> , 05/03/2026, 2026 European Conference of Radiology (ECR), organised by the ESR in Vienna (Austria).
ISMRM Iberian Chapter 2025	Moderator of oral scientific session <i>"Clinical session"</i> , 03/07/2025, 2025 annual meeting of the Iberian Chapter of the International Society for Magnetic Resonance in Medicine (ISMRM), Barcelona (Spain).
ISMRM 2022	Moderator of oral "power pitch" scientific session <i>"Motion correction"</i> , 10/05/2022, 2022 annual meeting of the International Society for Magnetic Resonance in Medicine (ISMRM), London (UK).
ISMRM 2021b	Moderator of oral scientific session <i>"Microstructure: Modelling Gray & White Matter Diffusion"</i> , 19/05/2021, 2021 virtual annual meeting of the ISMRM.
ISMRM 2021a	Facilitator of poster session <i>"Diffusion Applications: Brain & Spine"</i> , 18/05/2021, 2021 virtual annual meeting of the ISMRM.

PARTICIPATION IN EVALUATION PANELS

<i>Research Council of Finland 2026</i>	Grant reviewer for the Research Council of Finland, 2026 Call for Clinical Medicine Research (online evaluation).
<i>PhD thesis — UAB 2026</i>	PhD thesis reviewer for the Universitat Autònoma de Barcelona (UAB), Cerdanyola/Barcelona (Spain). Thesis title: “ <i>New immunoPET radiopharmaceuticals for immune cell assessment in ulcerative colitis</i> ”. Evaluation report submitted on: 04/08/2026 (online). Role: external expert reviewer (before thesis defense). Degree: PhD in Medicine.
<i>MSc thesis — UAB 2026</i>	MSc defense examiner at the Universitat Autònoma de Barcelona (UAB), Cerdanyola/Barcelona (Spain). Thesis title: “ <i>Metabolic profiling and classification of brain tumours via cNMF analysis of MRS data</i> ”. Exam date: 14/07/2026 (in person at UAB). Role: defense examiner. Degree: MSc in Bioinformatics.
<i>Grants — Finland 2025</i>	Grant reviewer for the Research Council of Finland, 2025 Call for Clinical Research (online evaluation).
<i>PhD thesis — EPFL 2024</i>	PhD defense at the École Polytechnique Fédérale de Lausanne (EPFL), Lausanne (Switzerland). Thesis title: “ <i>Lymph node microstructure imaging using diffusion MRI: From simulations to acquisitions</i> ”. Exam date: 08/10/2024 (in person in Lausanne). Role: external defense examiner. Degree: PhD. Department: Center for Biomedical Imaging (CIBM), EPFL. Thesis link here .

ORGANISATION OF SCIENTIFIC EVENTS

<i>UB Seminar 2026</i>	Institute of Neuroscience Seminar at <i>Universitat de Barcelona 2026</i> : “ <i>Quantifying blood-brain barrier permeability using MRI</i> ”, 24/04/2026. Organisers and moderators: Roser Sala-Llonch, Francesco Grussu.
<i>ISMRM Iberian Chapter 2025</i>	ISMRM Iberian Chapter: 2025 annual meeting of the Iberian Chapter of the International Society for Magnetic Resonance in Medicine (ISMRM) (Barcelona, scheduled for 3-4/07/2025). Organisers: Marco Rius I (IBEC), Ortega Machuca MA (IBEC), Candiota AP (UAB), Julià-Sapé M (UAB), Muñoz Moreno E (IDIBAPS), Lope-Piedrafita S (UAB), Barba Vert I (UVic), Benito M (UCM), Grussu F (VHIO), Gómez Cabeza D (IBEC).
<i>MIS ISMRM 2021</i>	Member Initiated Symposium at ISMRM 2020: “ <i>Looking Beyond Axons: Imaging the Immune System in White Matter</i> ”, 19/05/2021. Organisers: Cohen-Adad J, Grussu F, Kolind S.
<i>WMSG ISMRM 2019</i>	ISMRM White Matter Study Group Virtual meeting: “ <i>Myelin Imaging in the Spinal Cord at High Field</i> ”, 27/06/2019, joint meeting with the High-field Study Group. Organiser: Grussu F. Chair: Cohen-Adad J.
<i>MIS ISMRM 2019</i>	Member Initiated Symposium at ISMRM 2019: “ <i>Completing the Circle: Moving Multi-Parametric Neuro MRI into Clinical Practice and Trials</i> ”, 15/05/2019. Organisers: Vrenken H, Cohen-Adad J, Grussu F.
<i>CDMRI 2018</i>	MICCAI Workshop: Computational Diffusion MRI (CDMRI) 2018 (Granada, Spain, 20/09/2018). Organisers: Bonet-Carne E (UCL), Grussu F (UCL), Ning L (Harvard), Sepehrband F (USC), Tax C (Cardiff University).
<i>MUSHAC Challenge 2018</i>	MICCAI Challenge: Multi-shell dMRI harmonisation and enhancement (MUSHAC, part of CDMRI 2018). Organisers: Bonet-Carne E (UCL), Grussu F (UCL), Ning L (Harvard), Sepehrband F (USC), Tax C (Cardiff University).
<i>CDMRI 2017</i>	MICCAI Workshop: Computational Diffusion MRI (CDMRI) 2017 (Quebec City, Canada, 10/09/2017). Organisers: Grussu F (UCL), Kaden E (UCL), Ning L (Harvard), Tax C (Cardiff University), Veraart J (NYU).
<i>Data Harmonisation Challenge 2017</i>	MICCAI Challenge: Diffusion MRI data harmonisation (part of CDMRI 2017).

Organisers: Grussu F (UCL), Kaden E (UCL), Ning L (Harvard), Tax C (Cardiff University), Veraart J (NYU).

*Spinal Cord Grey
Matter
Segmentation
Challenge 2016*

Challenge: *Grey Matter Segmentation: What's there and What's next?* (part of SC MRI Workshop 2016). **Organisers:** Cohen-Adad J, Prados F, Landman B, Wheeler-Kingshott C, Summers P, Dupont S, Yiannakas M, Smith S, Gergely D, DeLeener B, Grussu F.

SCIENTIFIC PEER REVIEWING

2016-present

Reviewer for Bipolar Disorders; Functional Neurology; Frontiers; Human Brain Mapping; Journal of Magnetic Resonance Imaging; Journal of Neuroscience Methods; Magnetic Resonance in Medicine; NeuroImage; Medical Image Analysis; IEEE Transactions on Medical Imaging; npj Precision Oncology; NMR in Biomedicine; Scientific Data; 2019, 2021, 2022 ISMRM annual meetings; 2021 ISMRM Iberian Chapter post-grad meeting; 2021, 2025 Iberian Chapter annual meeting; 2019, 2020 MICCAI workshop on Computational Diffusion MRI.

OFFICIAL STUDENT SUPERVISION — COMPLETED

*MSc, VHIO and
UAB 2025/26*

Project co-supervision: *"Non-Invasive Tumour Microstructure Estimation Using Diffusion MRI: A Multi- Cohort Evaluation of Histo- μ Sim"*. **Degree:** Master's in "Bioninformatics", Universitat Autònoma de Barcelona (UAB; Barcelona, Spain). Project carried out at VHIO, Barcelona (Spain). **Supervised in:** 01/02/2026-14/07/2026 (completed). **Final thesis mark:** 9.69/10.

*PhD, VHIO and
UB 2022-26*

Project co-supervision: *"A simulation framework for characterizing cancer microvasculature with histology-informed diffusion MRI"*. **Degree:** PhD programme in Biomedicine (research line in biomedical engineering), Universitat de Barcelona (UB; Barcelona, Spain). Project carried out at VHIO, Barcelona (Spain). **Supervised in:** 30/09/2022-23/04/2026 (completed). **Final mark:** excellent and *cum laude*; international PhD award.

*MEng, VHIO and
TU Delft 2022/23*

Project co-supervision: *"Unraveling tumour microstructure through diffusion MRI using histology-powered artificial intelligence"*. **Degree:** Master's in "Biomedical Engineering", Delft University of Technology (TU Delft), (Delft, The Netherlands). Project carried out at VHIO, Barcelona (Spain). **Supervised in:** 09-2022/06-2023 (completed).

*MSc, VHIO and
UI La Rioja
2022/23*

Project co-supervision: *"Prediction of advanced biomarkers from clinical diffusion Magnetic Resonance Imaging"*. **Degree:** Master's in "Artificial Intelligence", Universidad Internacional de La Rioja (Logroño, Spain). Project carried out at VHIO, Barcelona (Spain). **Supervised in:** 2022/2023 (completed).

*MRes, UCL
2018/19*

Project co-supervision: *"Improving the differential diagnosis between Neuromyelitis Optica Spectrum Disorder and Multiple Sclerosis using MRI"*. **Degree:** MRes in "Clinical Neuroscience", Institute of Neurology, University College London (UCL, London, UK). **Supervised in:** 2018/2019 (completed).

*MSc, UCL
2018/19*

Project co-supervision: *"Evaluation of quantitative MRI indices reproducibility across scanner upgrade"*. **Degree:** MSc in "Advanced neuroimaging", Institute of Neurology, UCL (London, UK). **Supervised in:** 2018/2019 (completed).

*MSc, UCL
2017/18*

Project co-supervision: *"Investigation of multi-component T₁ relaxation at 3 Tesla"*. **Degree:** MSc in "Advanced neuroimaging", Institute of Neurology, UCL (London, UK). **Supervised in:** 2017/2018 (completed).

*MSc, UCL
2015/16*

Project co-supervision: *"Evaluation of strategies for co-registration between quantitative and anatomical magnetic resonance images of the human spinal cord"*. **Degree:** MSc in "Advanced neuroimaging", Institute of Neurology, UCL (London, UK). **Supervised in:** 2015/2016 (completed).

OFFICIAL STUDENT SUPERVISION — ONGOING

PhD, VHIO and UB 2023-27	Project co-supervision: “Novel Magnetic Resonance Imaging biomarkers for precision medicine in oncology: integrating multi-omics and real-world data strategies”. Degree: PhD programme in Biomedicine (research line in biomedical engineering), Universitat de Barcelona (UB; Barcelona, Spain). Project carried out at VHIO, Barcelona (Spain). Supervised in: 2023-2027 (ongoing).
PhD, VHIO and UB 2022-26	Project co-supervision: “Histology-informed diffusion MRI simulations and artificial intelligence for cancer microstructure characterization”. Degree: PhD programme in Biomedicine (research line in biomedical engineering), Universitat de Barcelona (UB; Barcelona, Spain). Project carried out at the Vall d’Hebron Institute of Oncology (VHIO), Barcelona (Spain). Supervised in: 2022-2026 (ongoing).

TEACHING EXPERIENCE

UB Lecture	Lecture: “Advanced medical imaging: a journey through engineering, physics and computer science”. Subject/module: “Artificial intelligence in biomedical engineering”. Degree: BEng in “Biomedical engineering”, Universitat de Barcelona (Barcelona, Spain). Conveyed: 12/05/2026. Face-to-face teaching: 2 hours.
UVi-UCC Lecture and Workshop	Lecture and Workshop: “Medical image analysis”. Subject/module: “Interactomics”. Degree: MSc in “Omics Data Analysis”, Universitat de Vic-Universitat Central de Catalunya, UVic-UCC (Vic and Barcelona, Spain). Conveyed: 17/03/2026. Face-to-face teaching: 4 hours.
UCL Lecture	Lecture: “Image optimisation: SNR, CNR and sources of artifacts”. Degree: MSc in “Advanced neuroimaging”, Institute of Neurology, UCL (London, UK). Conveyed: 2019/20, 2017/18, 2016/17, 2015/16. Face-to-face teaching: 4 hours (1 hour per year). Additional tasks: exam preparation and assessment.
UCL Workshop	Workshop: Hands-on with a portable MRI scanner . Degree: MSc in “Advanced neuroimaging”, Institute of Neurology, UCL (London, UK). Conveyed: 2017/18, 2016/17, 2015/16. Face-to-face teaching: 12 hours (4 hour per year).
UniPV Workshop	Workshop: “Model fitting for quantitative MRI”. Degree: MEng in “Biomedical Engineering”, University of Pavia (Pavia, Italy). Conveyed: 2016/17. Face-to-face teaching: 2 hours. Additional tasks: exam preparation and assessment.
UCL Lecture	Lecture: “Magnetic resonance image formation”. Degree: MSc in “Advanced biomedical imaging”, Centre for Advanced Biomedical Imaging, UCL (London, UK). Conveyed: 2015/16. Face-to-face teaching: 1 hour.

OPEN SCIENCE

	Code released through GitHub (http://fragrussu.github.io and http://github.com/radiomicsgroup)
GitHub 2024c	Histo-μSim: histology-informed cancer diffusion MRI (link).
GitHub 2024b	SpinFlowSim: diffusion MRI simulator in vascular networks (link).
GitHub 2024a	BodyMRITools: python code for body diffusion MRI processing (link).
GitHub 2022	MChepato: Code and synthetic data for Grussu et al, MRM 2022 (link). Available in Zenodo as record 6645258, doi: 10.5281/zenodo.6645258 .
GitHub 2020b	qMRI-Net: MRI signal model fitting based on artificial intelligence (link).

GitHub 2020a	SARDU-Net : data-driven, model-free quantitative MRI protocol design (link).
GitHub 2019b	MyRelax : tools for myelin and relaxation MRI analyses (link to the latest version). Version 1.0.0 available in Zenodo as record 4561898, doi: 10.5281/zenodo.4561898 .
GitHub 2019a	MRITools : tools for handling and managing research MRI scans (link).
GitHub 2016	StructureTensorToolbox : tools for analysis of 2D histological images (link).

Data sets released through Zenodo

Zenodo 2024b	"Histology-informed microstructural diffusion simulations for MRI cancer characterisation (Histo- μ Sim): <i>ex vivo</i> mouse data". Grussu F, Grigoriou A, Macarro C and Perez-Lopez R. Data set. Zenodo record 14559355, 2024, doi: 10.5281/zenodo.14559355 .
Zenodo 2024a	"Histology-informed microstructural diffusion simulations for MRI cancer characterisation (Histo- μ Sim): <i>histology substrates</i> ". Grigoriou A, Macarro C, Perez-Lopez R and Grussu F. Data set. Zenodo record 14559103, 2024, doi: 10.5281/zenodo.14559103 .

DISSEMINATION TO THE WIDER PUBLIC

2024b	Lab demonstration: " <i>Uso de la inteligencia artificial en la investigación contra el cáncer</i> " (" <i>Using artificial intelligence in cancer research</i> ") at the 2024 European Researchers' Night (27/09/2024), CosmoCaixa, Barcelona (Spain).
2024a	Video: " <i>Resonancia magnética para combatir el cáncer</i> " (" <i>Magnetic Resonance Imaging to fight cancer</i> "), released through the YouTube channel of the Vall d'Hebron Institute of Oncology (Barcelona, Spain) (link , video in Spanish).
2022b	Science fair: " <i>Demostración de análisis avanzado de imágenes por resonancia magnética y datos co-localizados de microscopía</i> " (" <i>Demonstration of advanced analysis of MR images and co-localised microscopy</i> ") at the 2022 European Researchers' Night (30/09/2022), CosmoCaixa, Barcelona (Spain).
2022a	" <i>Innovating Magnetic Resonance Imaging to fight diseases</i> ". Università della Terza Età (UniTre), Mòguru, Italy (12/04/2021). Talk given to a general audience in Sardinian language.
2017	" <i>Why to get vaccinated and avoid misinformation: the scientific method in modern medicine</i> ". Event for a general audience in Italian and Sardinian, Mòguru, Italy (17/08/2017). Organisers: Grussu F, Tur C, Coccollone E, Broccia S.
2015	Participation at <i>MS Frontiers 2015</i> (29-30/06/2015), organised by the UK Multiple Sclerosis Society and bringing together researchers and Multiple Sclerosis patients.
2013	UCL stall at <i>Science Uncovered</i> , 2013 European Researcher's Night (28/09/2013), London Science Museum.

CONFERENCE & WORKSHOP PROCEEDINGS: FIRST AUTHORSHIP

ECR 2026	"Response assessment and prognosis prediction in cancer immunotherapy using multi-parametric MRI: <i>preliminary evidence</i> ". Grussu F et al. European Conference of Radiology (ECR) 2026, C-16681 (EPOS TM oral + digital poster presentation, 07/03/2026), doi: 10.26044/ecr2026/C-16681 .
ESMRMB 2024	"Histological interpretation of Susceptibility-Perturbation MRI in human tumours of the liver". Grussu F et al. Proc of the European Society for Magnetic Resonance in Medicine and Biology (ESMRMB) 2024 annual meeting (traditional poster presentation).

ISMRM 2024b	<i>"Histology-informed biophysical diffusion MRI model selection for enhanced liver cancer immunotherapy assessment"</i> . Grussu F et al. Proc of the International Society for Magnetic Resonance in Medicine (ISMRM) 2024 annual meeting (oral presentation).
ISMRM 2024a	<i>"Two-axon population (TAP) modelling for large axon diffusion imaging in the peripheral nervous system"</i> . Grussu F et al. Proc of ISMRM 2024 (oral presentation).
ISMRM Iberian 2023	<i>"Extra-cellular liver diffusion modelling at high b-value: a preclinical MRI-histology study"</i> . Grussu F et al. Iberian Chapter of the ISMRM 2023 (oral presentation).
ISMRM 2022b	<i>"Histological correlates of DR-HIGADOS microstructural metrics in the mouse and human liver"</i> . Grussu F et al. Proc of ISMRM 2022 (power-pitch presentation).
ISMRM 2022a	<i>"Inter-scanner reproducibility and variability assessment of advanced liver diffusion MRI metrics"</i> . Grussu F et al. Proc of ISMRM 2022 (d-poster presentation).
ISMRM 2021b	<i>"DR-HIGADOS: a new diffusion-relaxation framework for clinically feasible microstructural imaging of the liver"</i> . Grussu F et al. Proc of ISMRM 2021 (oral presentation, Magna cum Laude award).
ISMRM 2021a	<i>"Investigating the relationship between diffusion MRI signal cumulants and hepatocyte microstructure at fixed diffusion time"</i> . Grussu F et al. Proc of ISMRM 2021 (d-poster presentation).
ISMRM 2020	<i>"SARDU-Net: a new method for model-free, data-driven experiment design in quantitative MRI"</i> . Grussu F et al. Proc of ISMRM 2020 (power-pitch presentation, Magna cum Laude award).
ISMRM 2019	<i>"Clinically viable g-ratio imaging with unified readout at 3T: evaluation and comparison"</i> . Grussu F et al. Proc of ISMRM 2019 (e-poster presentation).
ISMRM 2018b	<i>"Magnitude versus complex-valued images for spinal cord diffusion MRI: which one is best?"</i> . Grussu F et al. Proc of ISMRM 2018 (oral presentation, Magna cum Laude award).
ISMRM 2018a	<i>"A unified signal readout improves denoising of multi-modal spinal cord MRI"</i> . Grussu F et al. Proc of ISMRM 2018 (poster presentation).
ISMRM 2017b	<i>"Origin of the time dependence of the diffusion-weighted signal in spinal cord white matter"</i> . Grussu F et al. Proc of ISMRM 2017 (oral presentation).
ISMRM 2017a	<i>"A unified signal readout for reproducible multimodal characterisation of brain microstructure"</i> . Grussu F et al. Proc of ISMRM 2017 (e-poster presentation, Magna cum Laude award, finalist at the Diffusion Study Group competition).
Brain School 2017	<i>"Whole-brain macromolecular tissue volume mapping: A comparison of imaging readouts at 3 Tesla"</i> . Grussu F et al. School of Brain Cells and Circuits "Camillo Golgi". published in Frontiers, ISBN 978-288945-584-3 (poster presentation).
ISMRM Scientific Workshop 2016	<i>"Optimal histological quantification of neurite orientation dispersion for the validation of diffusion MRI"</i> . Proc of ISMRM Scientific workshop "Breaking the barriers of diffusion MRI" (poster presentation + Power Pitch).
ISMRM 2016	<i>"Axon diameter distribution influences diffusion-derived axonal density estimation in the human spinal cord: in silico and in vivo evidence"</i> . Grussu F et al. ISMRM 2016 (poster presentation, finalist at the Diffusion Study Group competition).
ECTRIMS 2015	<i>"Quantitative histological validation of NODDI MRI indices of neurite morphology in multiple sclerosis spinal cord"</i> . Grussu F et al. European Committee for Research and Treatment of Multiple Sclerosis (ECTRIMS) 2015 (poster presentation, short-listed for poster prize competition).
MS Frontiers 2015	<i>"Histological correlates of NODDI in the multiple sclerosis spinal cord"</i> . Grussu F et

al. MS Frontiers 2015, annual scientific meeting of the UK Multiple Sclerosis Society (**oral and poster presentation**).

- ISMRM 2015b *"Quantitative histological correlates of NODDI orientation dispersion estimates in the human spinal cord"*. Grussu F et al. Proc of ISMRM 2015 (**oral presentation, Magna cum Laude** award).
- ISMRM 2015a *"Histological metrics confirm microstructural characteristics of NODDI indices in multiple sclerosis spinal cord"*. Grussu F et al. Proc of ISMRM 2015 (**oral presentation, Magna cum Laude** award).
- British Chapter of the ISMRM 2014 *"Characterisation of single-shell NODDI fitting in spinal cord grey and white matter"*. Grussu F et al. Proc of the British Chapter of the ISMRM 2014 (poster presentation).
- ISMRM 2014b *"Neurite orientation dispersion and density imaging of the cervical cord in vivo"*. Grussu F et al. Proc of ISMRM 2014 (poster presentation).
- ISMRM 2014a *"Single-shell diffusion MRI NODDI with in vivo cervical cord data"*. Grussu F et al. Proc of ISMRM 2014 (poster presentation).
- ISMRM Workshop 2013 *"In vivo estimation of neuronal orientation dispersion and density of the human spinal cord"*. ISMRM Scientific workshop "Multiple sclerosis as a whole-brain disease" (**oral presentation**).
- ISMRM 2013 *"Towards spinal cord microstructure mapping with the neurite orientation dispersion and density imaging"*. Grussu F et al. Proc of ISMRM 2013 (poster presentation, finalist at the White Matter Study Group poster competition).

CONFERENCE & WORKSHOP PROCEEDINGS: SENIOR AUTHORSHIP

- ESMRMB 2026c *"Towards the Acquisition-Independence of Microvascular Biomarkers in Diffusion MRI: a Feasibility Study at 0.55T"*. Voronova A, ..., and Grussu F. Proc of ESMRMB 2026, PG.226 (poster presentation).
- ESMRMB 2026b *"Non-invasive glioma grading through biophysical diffusion MRI modelling: the complementary role of Histo- μ Sim and SANDI"*. Monetta A, ..., and Grussu F. Proc of ESMRMB 2026, PG.400 (poster presentation).
- ESMRMB 2026a *" μ NN: a physics-informed histology-MR neural network for microstructural diffusion imaging in cancer"*. Grigoriou A, ..., and Grussu F. Proc of ESMRMB 2026, PG.031 (**oral presentation**).
- ISMRM Iberian Chapter 2025 *"Simplicity is a virtue: histology-informed model comparison selects simple diffusion representations in a colorectal cancer metastasis specimen"*. Grigoriou A, ..., and Grussu F. 2025 ISMRM Iberian Chapter annual meeting, p.49 (poster presentation). **2nd prize for the best poster presentation**.
- ISMRM Workshop 2025b *"Which Microvascular Properties Can We Probe in Clinical Settings with Diffusion MRI?"*. Voronova A, ..., and Grussu F. 2025 ISMRM Workshop on 40 Years of Diffusion: Past, Present and Future Perspectives (**oral presentation**).
- ISMRM Workshop 2025b *"Histology-Informed Microstructural Diffusion Simulations (Histo- μ Sim) for Enhanced Diffusion MRI Parameter Estimation in Cancer"*. Grigoriou A, ..., and Grussu F. 2025 ISMRM Workshop on 40 Years of Diffusion: Past, Present and Future Perspectives (power-pitch presentation), p. 202.
- ESMRMB 2024b *"Biologically-realistic blood flow simulations reveal complex features of vascular IVIM signals"*. Voronova A, ..., and Grussu F. Proc of ESMRMB 2024 (traditional poster presentation).
- ESMRMB 2024a *"Histology-informed cell size distribution mapping with diffusion MRI"*. Grigoriou A, ..., and Grussu F. Proc of ESMRMB 2024 (traditional poster presentation).

ISMRM 2024b	<i>"FlowSim: a blood flow simulator for histology-informed diffusion MRI micro-vasculature mapping in cancer"</i> . Voronova A, ..., and Grussu F. Proc of ISMRM 2024 (power-pitch presentation, Summa cum Laude award, Diffusion Study Group prize, best application (poster category)).
ISMRM 2024a	<i>"A Monte Carlo simulation framework for histology-informed diffusion MRI parameter estimation in cancer"</i> . Grigoriou A, ..., and Grussu F. Proc of ISMRM 2024 (power-pitch presentation, Magna cum Laude award).
ISMRM Scientific Workshop 2022	<i>"A systematic comparison of machine learning approaches for diffusion-relaxation MRI protocol enhancement in advanced solid tumours"</i> . Macarro C, ..., and Grussu F. ISMRM Workshop on Diffusion MRI: From Research to Clinic 2022 (poster presentation).
ISMRM 2019	<i>"Sensitivity of NODDI and two-compartment SMT parameter maps in multiple sclerosis"</i> . Johnson D, ..., and Grussu F. Proc of ISMRM 2019 (e-poster presentation).

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